

Abstract

Objectives: Colorectal cancer (CRC), the third most common cancer in the USA, is a leading cause of cancer-related death worldwide. Up to 60% of patients develop liver metastasis(CRLM). Treatments like radiation and ablation therapies require disease segmentation for planning and therapy delivery. For ablation, ablation-zone segmentation is required to evaluate disease coverage.

We hypothesize fully convolutional (FC) neural networks, trained using novel methods, will provide rapid and accurate identification and segmentation of CRLM and ablation-zones.

Methods: Four FC model styles were investigated: Standard-, Residual-, Dense-3D-UNet, and Hybrid-WNet. Models were trained on 92 patients from the Liver Tumor Segmentation(LiTS) challenge. For the evaluation, we acquired 15 patients from the 3D-IRCADb database, 18 patients from our institution(CRLM=24, ablation-zone=19), and submitted to the LiTS challenge(n=70). Qualitative evaluations of our institutional data were performed by two board certified radiologists(interventional, diagnostic), and a radiology-trained physician fellow, using a Likert scale 1-5.

Results: The most accurate model was the Hybrid-WNet. On a patient-by-patient basis in the 3D-IRCADb dataset, the median(min-max) Dice similarity coefficient(DSC) was 0.73(0.41-0.88), median surface distance was 1.75mm(0.57–7.63mm), and number of false-positives was 1(0-4). In the LiTS challenge(n=70), the global DSC was 0.810. The model sensitivity was 98%(47/48) for sites ≥ 15 mm in diameter. Qualitatively, 100%(24/24; minority vote) of the CRLM and 84%(16/19; majority vote) of the ablation-zones had Likert scores ≥ 4 .

Conclusion: The Hybrid-WNet model provided fast(<30 seconds) and accurate segmentations of CRLM and ablation-zones on contrast-enhanced CT scans, with positive physician reviews.

Introduction

Colorectal Cancer in the United States

Colorectal cancer (CRC) is the third most common cancer in the United States in both men and women¹ and a leading cause of cancer-related death worldwide². The main cause of death for CRC patients is due to metastasis⁴. Up to 60% of patients develop colorectal liver metastasis (CRLM) over the course of their disease, with 25% presenting with CRLM at diagnosis³. Such facts highlight the importance of liver-directed loco-regional therapies (LRT) for these patients.

While several treatment options are available for CRLM (in particular radiation and ablation therapies), they all rely on accurate estimation of disease extent, usually involving cross-sectional imaging with contrast-enhanced CT (CECT) or MRI. CRLM often appears as hypo-enhancing lesions on routine CECT portal-venous phase images. However, their detection can be challenging owing to ill-defined margins, particularly for sub-centimeter lesions.

Both radiofrequency and microwave ablation interventions aim for a minimum margin to be achieved around the disease to ensure that all microscopic disease is treated. This requires both segmentation of disease on pre-treatment images, and the ablation zone on post-treatment images to assess the ablation margin⁵. The ablation zone is hypo-enhanced on CECT images, similar to the CRLM. A clinical trial is underway (Identifier: NCT04083378) to map the CRLM from pre-treatment to post-treatment imaging and assess treatment efficacy, but manual segmentations of both the disease and the ablation zone is still required⁶, adding time to the procedure.

To date, automated liver disease segmentation tasks have either largely focused on primary liver disease, or have not included qualitative evaluation of generated contours^{7–10}. New institution- and society-sponsored competitions, such as the Liver Tumor segmentation (LiTS) challenge¹¹ and the 3D-IRCADb01⁷ dataset have included data from both primary and secondary liver cancers, enabling investigation, development and comparison of automatic segmentation algorithms using public data.

We hypothesized that fully convolutional neural networks, trained using novel methods to account for the challenges of varying disease size, will provide rapid and accurate identification and segmentation of both CRLM and ablation cavities. We believe that this approach will facilitate the automated detection of CRLM, radiation treatment planning for CRLM, and the evaluations of margin in ablation therapy.

Materials and Methods

Quantitative Training, Validation, and Testing

To ensure reproducibility by other institutions, we used data provided by the publicly available LiTS challenge¹¹. LiTS consists of CECT scans from 131 patients with primary and secondary liver disease. Ground truth segmentations of the liver and disease were provided. Each image was reviewed by XXX to remove the data showing hyper-enhancing metastases, or lacking image acquisition parameters. 92 patients remained.

We evaluated our model via submission to the LiTS challenge, and the 3D-IRCADb01 publically available dataset¹². Table 1 shows the image acquisition parameters for the training, validation, and test sets. Five of the patients from the 3D-IRCADb01 dataset were excluded: patients 5, 11, and 20 had no CRLM,

patient 12 had a large cystic lesion at the base of the liver, and patient 18 had CRLM with atypical enhancement pattern.

Data Pre-Processing

Image Intensity Manipulation

We calculated a patient-specific mean and standard deviation for normalization on the basis of the Full-Width-Half-Max of the values within the liver, this reduced outliers as compared to using a global mean and standard deviation. The image intensity outside of the masked liver was set to be equal to 0.

Voxel Size Resampling

We resampled all training and validation images and ground-truth segmentation to 1mm slice thickness using bi-linear interpolation.

Training Image “Slabs”

To account for disparities in class representation, where smaller structures are inherently “worth less” than large structures, we distributed our training into unique “slabs”. Simple class weighting would not solve this problem as it would result in the model weighing cases with extensive disease cases as more important than the less extensive disease cases. Each independent disease site was divided into “slabs” of 32x120x120 voxels. Figure 1 illustrates several disease slabs for one patient. After extraction, the training dataset consisted of 572 unique samples.

The validation and test set were not broken into slabs, with the entire liver being passed at once for evaluation and testing.

Architectures

Four architectures were investigated: Standard 3D-UNet, Residual 3D-UNet, Dense 3D-UNet, and Hybrid-WNet (pre-trained Standard 2D-UNet with a 3D-DenseUNet). The basic framework remained the same for the Standard, Residual, and Dense 3D-UNets (Figure 2A). The differences in the Standard, Residual, and Dense 3D-UNets are represented in Figure 2B. The Hybrid-WNet architecture is shown in Figure 3. A list of parameters for each architecture is listed in Table 2.

Residual 3D-UNet

The Standard 3D-UNet was expanded to include residual connections for each convolution block in a layer, with motivation from the ‘ResNet’ architecture¹³. Residual connections have the benefit of allowing a “flow” of loss from previous convolutions. This allows the model to create skip-connections over convolutions that might not be necessary.

Dense 3D-UNet

A more complete “flow” of loss from previous convolutions can be realized with the DenseNet architecture¹⁴. This architectural style allows previous convolutions to be re-used. The reuse of previous convolutions allows the number of filters to be significantly reduced, the increase in total number of filters is referred to here as the “growth rate”.

Hybrid-WNet

The architecture combines 2D features extracted from the pre-trained 2D DenseNet-121¹⁴ in Tensorflow¹⁵ with a 3D convolutional neural network. We coined the term Hybrid-WNet on the basis of the W-shaped appearance of two UNets beside each other (Figure 3).

Our Hybrid-WNet architectural style was inspired by Li et al.⁹ with substantial alterations. First, in architecture training, we broke the training process into four steps: 1) training only the new decoding side of the DenseNet 121, 2) training the entire DenseNet121, 3) training only the 3D network with the extracted 2D features, and 4) entire end-to-end training. By breaking up the training process in this fashion, we ensured that we could use high learning rates without the risk of “untraining” pretrained layers. Second, our 3D DenseNet contained truly dense layers, with extracted features shared throughout the entirety of each layer; this enabled the use of significantly fewer features.

Model Training

All model training was performed using NVIDIA-Tesla V100 32GB GPUs¹⁶. All model creation, training, optimization, and evaluation was performed using Tensorflow2.2.0¹⁷. Models were optimized using a sparse categorical cross entropy loss and Adam optimizer. Mixed precision was enabled to reduce the training time.

The model was trained with two inputs: CT image and binary mask of the liver. The mask automatically assigns a background to any voxel outside the liver. Training involved passing $B*N*H*W*C$ tensors to the model, where the (B)atch varied from 8, to 16, the (N)umber of slices was 32, the (H)eight was 120, the (W)idth was 120, and the (C)hannels was 2 (image and liver mask). Thus, a single pass might be $8*32*120*120*2$ in size.

Training the DenseNet121 UNet

When training the 2D aspect of the Hybrid-WNet, 3D slabs were reshaped into stacked 2D images. For example, a batch of $8*32*120*120*1$ would be transformed into $8*32*120*120*1$. In the first training iteration, all weights on the pre-trained encoding architecture were frozen. Next, all weights were made trainable, allowing the model to tweak any pre-trained layers.

Training the Combined 2D-3D WNet

After training the 2D part of the W-Net, we concatenated the 2D features to the input of the 3D model. All weights from the 2D network were initially frozen, and only the 3D model trained. Next, all weights were unfrozen, allowing the model to be fine-tuned.

Hyper-Parameters

For training each model, a variation in the cyclical learning rate¹⁸ was used (GitHub link: anonymized for review), with linear increase and decrease between min and max. Optimal learning rates vary based on each architecture parameter, with the minimum and maximum learning rates identified using an in-house function (Githb link: anonymized for review), Supplementary Figure 1.

Model Optimization

We ran each model three times using randomly initialized variables to reduce the likelihood of poor initialization. Plotnine and Tensorflow’s Tensorboard (<https://github.com/tensorflow/tensorboard>) was used to identify trends and direct model training. The final model was selected on the basis of the Dice similarity coefficient (DSC) between the validation set and the ground truth.

Prediction images were visualized during training to assess the training process (Figure 4) as a Tensorflow callback (Githb link: anonymized for review).

Model Evaluation

Seed and Threshold Values

The disease predictions from our model ranged from 0-1. The most inferior and superior aspects of a disease site often had a lower probability than the center of the disease. For this reason, we investigated seed-point growth for the final prediction. Seed points were created to define the likely starting point of a disease site, and then grown outwards to a threshold value. We investigated seed values from 0.25-0.95 in 0.01 increments, and threshold values from 0.2 to 0.8 in 0.01 increments.

Quantitative Evaluation

Model performance was evaluated on the test set using the (DSC) and Median Surface Distance between the manual and predicted segmentations. We report on predictions in a disease site-by-site, patient-by-patient, and “global” basis.

For the site-by-site comparison, each non-connected disease segmentation of the test patients was considered an independent case. We computed the metrics between the manual segmentation and the closest continuous predicted disease segmentation, using the distance between site centroids. For the patient-by-patient comparison, we computed the metrics between the manual and predicted segmentations within the entire liver. For global DSC, all images were stacked together, this metric comes from the LiTs challenge¹¹. **All metrics were computed using the original image resolution.**

The sensitivity of the model was evaluated on a site-by-site basis, where disease was considered identified if at least 45% of the ground truth overlapped with prediction. False-positive volume was defined as the volume of the predicted segmentation outside of the ground truth segmentation and was comprised of two errors: over-segmentation and erroneous segmentation. Erroneous false-positive volume was quantified as the volume that was completely unconnected from any ground truth segmentations and the over-segmentation false volume as the total predicted volume minus the erroneous volume (Figure 5).

Qualitative Evaluation

For large structures, a high DSC can hide clinically important inaccuracies, while for smaller structures, a low DSC can be overly critical¹⁹. Our qualitative assessment of both CRLM and ablation segmentations was performed by a radiology-trained physician fellow (1, XXX) and two board-certified radiologists (2, XXX), an interventional radiologist, and (3, XXX), a diagnostic radiologist, both of whom have more than 10 years of experience.

We retrospectively identified 18 patients, who had previously undergone targeted thermal ablation therapy at our institution under an Institutional Review Board-approved study (IRB: 2019-0213); these patients had 24 CRLM sites and 19 ablation sites. The predicted CRLM and ablation contours were scored independently based on a Likert quality scale of 1-5 (Supplementary Table 1).

Results

The best validation loss scores were, Standard UNet: 0.041, Residual UNet: 0.024, DenseUNet: 0.016, DenseNet2D (Encoder frozen): 0.022, DenseNet2D (All trainable): 0.013, DenseNet3D (2D frozen): 0.011, DenseNet3D (All trainable): 0.0092.

Quantitative

The best model was the Hybrid-WNet model: the 3D-UNet contained two layers, two convolution blocks, and 32 filters and had a convolution lambda of two. The model consisted of 14,497,600 parameters (14,408,960 trainable and 88,640 non-trainable).

A seed value of 0.67 and threshold value of 0.30 resulted in the highest overall DSC in the validation dataset. LiTs test set (n=70) predictions required a mean of 9.58 seconds, with a standard deviation of 2.32 seconds.

Site-by-Site evaluation

3D-IRCADb

On a site-by-site basis, the median surface distance, DSC, and sensitivity are presented in Table 3. Sites are distributed into two groups based on diameter: sites < 15 mm and ≥ 15 mm. For sites ≥ 15 mm (n=48), mean DSC was 0.74 and sensitivity was 98%. For sites < 15 mm (n=73), mean DSC was 0.16 and sensitivity was 23%.

Institutional Data

For the pre-treatment CECT CRLM target disease (n=19 sites), the mean (min-max) DSC was 0.80 (0.59-0.91), with 84% (16/19) having DSC ≥ 0.76 . For the post-treatment CECT target ablation zones (n=14 sites), the mean (min-max) DSC was 0.75 (0.09-0.90), with 71% (10/14) having DSC ≥ 0.76 .

Patient-by-Patient evaluation

Patient-by-patient evaluation for 3D-IRCADb and institutional data are summarized in Table 4.

3D-IRCADb

Quantitative metrics on a patient-by-patient basis for 3D-IRCADb are summarized in the top of Table 4. Median DSC was 0.74, median surface distance was 1.95 mm, and median false-positive discoveries per patient was 1. Note that a single patient might have multiple disease sites of varying sizes. The global DSC score was 0.81.

Institutional Data

Quantitative metrics for the institutional data are summarized in the middle and bottom of Table 4. For the pre-treatment CECT CRLM target disease (n=15 patients), the median DSC was 0.78, median surface distance was 0.78 mm, and false-positive discoveries per patient was 1. For the post-treatment CECT target ablation zones (n=9 patients), the median DSC was 0.79, median surface distance was 0.76 mm, and false-positive discoveries per patient was 2.

Qualitative Evaluation

Institutional Data

Supplementary Table 2 shows the Likert scores of the two radiologists and radiology trained physician fellow for each targeted disease and ablation zone. The majority vote mean (min-max) Likert scores for the target disease volumes (n=24) were 4.8 (4-5). The majority vote mean (min-max) Likert scores for targeted ablation volume (n=19) were 4.1(2-5). All (24/24) of the CRLM contours and 84% (16/19) of the ablation contours had a majority Likert score ≥ 4 . Figure 6 shows the scores of the two radiologists and physician fellow on the CRLM and ablation zones.

Discussion

While fully convolutional networks have been previously investigated for the segmentation of CRLM⁸, only a quantitative assessment of model performance has been reported. A comparison of our model results to those of other authors is given in Table 5; our size distributions were varied to match those of previous work, and compared in both a site-by-site basis and patient-by-patient basis. While our model performed better in larger diameter sites (≥ 15 mm) by DSC and sensitivity compared to Vorontsov et al⁸, it performed more poorly with sites < 10 mm. Similar results have also been reported by Christ et al;⁷ however, their work focused on primary liver disease.

Within our institutional data, 84% (16/19) of the disease sites had a DSC >0.76 , which has been reported as the inter-observer variability for CRLM segmentation²⁰.

In this study, we proposed a Hybrid-WNet model architecture for the segmentation of the disease sites and ablation areas in the context of CRLM treated with ablation therapy. The model was further evaluated using the Likert scoring method by two board certified radiologists(XXX, interventional radiologist and XXX, diagnostic radiologist) and a radiology trained physician fellow(XXXs), with majority voting scores ≥ 4 out of 5 for 100%(24/24) of the disease and 84%(16/19) of the ablation segmentations. The prediction process has been implemented in a treatment planning system (RayStation 9B, RaySearch Laboratories, Sweden)²¹ and can perform segmentations in < 30 seconds, making it suitable for clinical use.

Our work is innovative because of the Hybrid-WNet model architecture and the training of the model; the model had with 98% (47/48) sensitivity on disease sites ≥ 15 mm, with predictions in < 30 seconds.

The proposed Hybrid-WNet architecture in this study differs from the model which inspired it⁹ by adding additions to the 2D feature extractor, and connecting all of the convolutional layers within the convolution blocks between the encoding and decoding of the 3D-DenseNet. Previously published studies proposed similar architectures that were focused only on dense connections within a single convolutional block^{22–24}. Our proposed implementation allows the model to receive inputs from every convolution of the same image size across the entire network. This global passing of convolution layers across the network removes bottlenecks in each convolution block, something that is particularly important with dense connections where convolutions can be re-used. The proposed model will be publicly available (Github: redacted for anonymization).

Segmentation of CRLM has historically been difficult because of the relatively small size of lesions and the large search area. The extent of disease can vary from patient to patient, from a single lesion to multiple lesions. Simple class weighting of disease would lead to favoring training in patients with more disease sites. Our proposed method of splitting the liver into slabs that were centered specifically on disease sites ensures that the model learns using a more balanced representation of data. The training workflow was designed to allow the model to learn from batches that contains cubic images from multiple patients at once, enabling the creation of a more generalized model.

Limitations

Our study has several limitations. The mean sensitivity was only 7% in disease sites < 10 mm in diameter compared with 98% in sites > 15 mm. We believe that the low sensitivity in small sites may be partly due to the test data used, where several 10-15-mm disease sites were present on a single scan slice.

Unfortunately, manual contours were not present for all of the institutional CRLM and ablation data, limiting our quantitative comparison to 19 of the 24 CLRM sites and 9 of the 19 ablation sites.

Majority voting showed poor Likert score (<4) for three ablation sites. The predictions for these sites can be seen in the Supplementary Figure 2, where disagreement about the boundary of the ablation zone and the time needed for correction resulted in a range of scores from the reviewers. We believe that suboptimal imaging quality during porto-venous CT acquisition phase might have negatively impacted ablation zone boundary identification in such patients. Optimizing imaging acquisition protocol intra-procedurally during ablation interventions might overcome this limitation.

The measurement of false-positive volume seemed to be highly biased when there were small amounts of over-segmentation on large tumors; this was the rationale for the creation of the erroneous and over-segmentation false-positive volumes. To ensure transparency, we included all three and relied on qualitative assessment to add weight to the quality of the contours.

Conclusions

The proposed Hybrid-WNet model provided fast (< 30 seconds) and accurate CRLM and ablation zone segmentations for CECT. Our model's results were well accepted by the reviewers, with all three scoring the disease segmentation (n=24) as 4 or higher on the Likert scale, and 84% (16/19) as 4 or higher with ablation segmentation. We believe that this model can provide clinical benefits in the detection of CRLM, the assessment of ablation therapy, and automated planning for radiation therapy.

Our proposed Hybrid-WNet provided fast (< 30 seconds) and accurate segmentation of colorectal liver metastasis and ablation zones, with largely positive physician reviews.

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Table and Figure Legends

Table 1: Image Acquisition Parameters of the LiTS Challenge for the Training, Validation, and Test Sets

Table 2: Investigated architectural hyper parameters for each architectural style. Note that the number of filters doubled after each pooling. 'Growth Rate' is unique to the Dense network.

Table 3: Mean, Min, and Max Dice Similarity Coefficient and Median Surface Distance and Sensitivity for Individual Disease Sites by Differing Size Criteria of the 3D-IRCADb Dataset

Table 4: Metrics of Dice Similarity Coefficient, Median Surface Distance (mm), False-Positive Discoveries (Per Patient), and False-Positive Volume (cc) for 15 3D-IRCADb Test Patients

Table 5: Comparison of CRLM Segmentation Results from Our Model and the Literature. Green Box Indicates More Disease Sites (n), Better Dice Similarity Coefficient (DSC), or Higher Sensitivity.

Supplementary Table 1: Likert Scoring for Assessment of Colorectal Liver Metastasis (CRLM) and Ablation Contours

Supplementary Table 2: Breakdown of Likert Score (1-5) for Predicted Contours of CLM and Ablation Sites by Each Reviewer

Figure 1: Liver distributed into individual slabs of 32x120x120. Disease was labeled as disease, regardless of the center of the slab. The validation and test set were not broken into slabs; our architectures accepted variable input sizes, with the entire liver being passed at once for evaluation and testing.

Figure 2: Top (A): Basic architecture framework. Bottom (B): Difference in convolution blocks, surrounded by green to indicate the same region in (A). Standard: previous feature maps are convolved and activated. Residual: previous feature maps are directly added to convolutional output in a skip-connection before activation. Dense: previous feature maps are continually concatenated together before activation and convolution. BN: batch normalization.

Figure 3: Hybrid-WNet. (Left) Pre-trained 2D DenseNet 121 converted into UNet, where final features were concatenated into 3D DenseNet. (Right) 3D DenseNet architecture was defined as two layers, two convolution blocks in layer 0, three convolution blocks in layer 1, and eight initial filters. BN: Batch Normalization

Figure 4: Example of visualization of prediction on validation data during training. Top row: Image being fed into the model, masked by the liver segmentation. Middle row: Ground truth of disease contours. Bottom row: Prediction of disease contours, set with a threshold of 0.5 to the binary mask.

Figure 5: (Left) Overlay between predicted (white) and ground truth (red) disease segmentations. (Right) Subtraction of the predicted and ground truth disease segmentation and over-segmentation of the disease volume (blue) and unconnected erroneous false-positive region (green). The sum of blue and green is a false-positive volume.

Figure 6: (Top) Likert score by each reviewer for each disease site. (Bottom) Likert score by each reviewer for each ablation site. A higher value is associated with higher quality.

Supplementary Figure 1: Learning rate finder figure. The learning rate is gradually increased from $10e-7$ to $10e-0$; the point where loss begins to decrease is the minimum learning rate (min LR), and the point where it begins to increase is the maximum learning rate (max LR).

Supplementary Figure 2: Three cases where majority voting had a Likert Score less than 4. (Top) We believe that the top failed due to similarity in appearance to the gallbladder. (Middle, Bottom) Model over-segmented regions around the ablation zones.